

2(a)(i)	red blood cell ; A erythrocyte / red blood corpuscle	1
2(a)(ii)	<i>any valid difference one mark</i> fluid R / tissue fluid, has, one from fewer (named) proteins / no large proteins / no plasma proteins ; A no named plasma proteins (albumen / fibrinogen) A fewer plasma proteins only if stated, they are small / leave the blood fewer / no, white blood cells ; A more macrophages no red blood cells ; no platelets ; <i>explanation to match difference 2 marks</i> (red blood cells, platelets, proteins, white blood cells) too large ; (fewer white blood cells) ref to some, white blood cells / phagocytes / macrophages monocytes, are able to move through ; cannot be, filtered / fit, through gaps / AW, between endothelial cells / in endothelial wall / in capillary wall ; idea that not filtered out under pressure from, plasma / blood ; or fluid R / tissue fluid, has one from higher partial pressure / AW, of, carbon dioxide ; A higher concentration / more <i>explanation to match difference 2 marks</i> produced by / waste of, (surrounding respiring) cells ; A (waste) product of respiration diffuses / AW, from cells to, (fluid) R / tissue fluid (before entering blood) ; idea that blood has come from lungs, where carbon dioxide has been excreted ; AVP ; e.g. diffusion through cells in capillary wall	3

Question	Answer	Marks
2(a)(ii)	or <i>fluid R / tissue fluid, has one from</i> lower partial pressure / AW, of, oxygen ; A lower concentration / less <i>explanation to match difference 2 marks</i> carried by the blood (from the lungs) ; A blood is more oxygenated diffuses / AW, from, red blood cells / Q / blood ; idea that filtered out under pressure from, plasma / blood ; AVP ; e.g. diffusion through cells in capillary wall or <i>fluid R / tissue fluid, has</i> higher concentration of, glucose / amino acids / fatty acids ; A more <i>explanation</i> carried by the blood to the respiring cells ; small ; can be, filtered / fit, through gaps / AW, between endothelial cells / in endothelial wall / in capillary wall ; idea that filtered out under pressure from, plasma / blood ; AVP ; e.g. diffusion through cells in capillary wall	
2(b)(i)	bilayer drawn with heads pointed outwards and tails pointed inwards ; phospholipid drawn with one (phosphate) head and two fatty acid tails ;	2
2(b)(ii)	<i>any one from:</i> cholesterol, maintains / regulates, the fluidity of the membrane ; A reduces fluidity at high temperature / increases the fluidity at low temperatures maintains / regulates, (mechanical) stability of the membrane / AW; prevents entry of hydrophilic substances / AW / ions ;	1

Question	Answer	Marks
2(c)(i)	<i>any two from:</i> increases heart rate ; increases blood pressure ; makes platelets stick together / makes blood more likely to clot ; damage to, endothelium / endothelial wall / capillary wall ; reduces blood flow to extremities / ref. vasoconstriction ;	2
2(c)(ii)	<i>any three from:</i> <i>similarities max 2</i> both have a (co-)transport protein ; both involve H⁺ / hydrogen ions / protons ; H⁺ move down the concentration gradient / by facilitated diffusion ; (sucrose and nicotine both) move against the concentration gradient ; <i>differences max 2</i> in nicotine transport, H⁺ ions and nicotine, move in opposite directions ; ora for sucrose transport H⁺ ions move from, the apoplast / cell wall, in sucrose transport (from the cytoplasm in nicotine transport) ; AVP ; e.g. ref to ATP	3